CLASSIFICATION OF WHITE BLOOD CELLS USING FEATURE EXTEACTION AND CELL SEGMENTATION

A PROJECT REPORT

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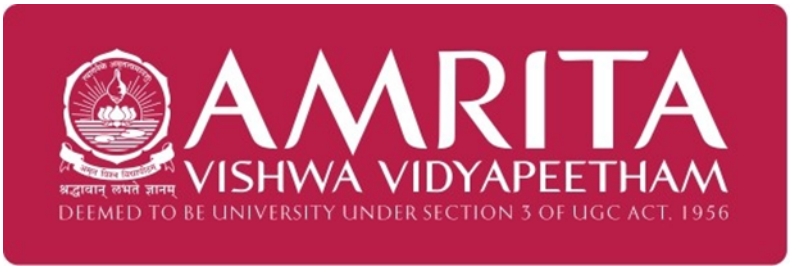
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BONAFIDE CERTIFICATE

This is to certify that this project report entitled **"CLASSIFICATION OF WHITE BLOOD CELLS USING FEATURE EXTEACTION AND CELL SEGMENTATION"** is the bonafide work of **"Avanindraa Nagarajan (Reg. No. CH.SC.U4AIE23004), Sai Raman D (Reg. No. CH.SC.U4AIE23010), Jothi Krishna K (Reg. No. CH.SC.U4AIE23022), Kishore N S (Reg. No. CH.SC.U4AIE23026), Himanshu M (Reg. No. CH.SC.U4AIE23028)"** who carried out the project work under my supervision as a part of End semester project for the course 22BIO211 - Intelligence of Biological Systems 2 .

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ABBREVIATIONS

WBC White Blood Cells

VGG Visual Geomentry Group

RBC Red Blood Cells

GBM Gradient Boost Machine

SMOTE Synthetic Minority Over-sampling Technique

PCR Polymerase Chain Reaction

CTS Computed Tomography Scan

GAN Generative Adversarial Network

ABSTRACT

Automated haematological analysis relies on white blood cell (WBC) segmentation to help diagnose diseases and plan treatments. In contrast to other traditional works, this one suggested a novel deep learning framework for the classification of white blood cells, which can have a significant influence on the identification of morphological cells in different cell types. In the suggested method, images are pre-processed and mapped with the appropriate labels. From the original image, we first extract the white blood cell using masking techniques. The segmented image is fed into the VGG16 model, which analyses it and provides a clear analysis of its features, including spatial arrangement and high level features. However, it is a hybrid dual-feature strategy that also calculates handcrafted features like Area, Perimeter, Solidity, Aspect Ratio, and Extent. We use LightGBM to assign a specific cell class to the processed image. The accuracy score and classification report are used to assess the model, and classification metrics have been developed. This suggested framework has made it easier and more efficient to identify and categories cell architecture. The goal of this work is to help pathologists and researchers identify white blood cells accurately and efficiently for quick diagnostics and a hygienic medical workflow.

Keywords: White blood cells, Classification, Contour segmentation, Feature Extraction, VGG16, LightGBM

CHAPTER 1

INTRODUCTION

Nuclei segmentation is a critical area in the medical image analysis field as it is an integral step in procedures such as disease diagnosis, tissue classification, and drug discovery. Though being very crucial, nuclei segmentation is extremely challenging due to the intricate nuclei morphology, the overlapping of nuclei, and artifacts in micrographic images. White blood cells(leukocytes) are crucial in the immune system and assist in repelling infection and disease. There are various kinds of WBCs like neutrophils, lymphocyte, monocytes, eosinophils, and basophils, and proper classification of each of these subtypes is of extreme importance in the diagnosis of hematological disorders such as leukemia, auto-immunologic diseases and infections. This research overcomes these drawbacks using a new hybrid approach, which incorporates the methods of Contour detection for nuclei segmentation and feature extraction and VGG16 and LightGBM for accurate WBC classification.

The approach suggested begins with pre-processing operations to normalize and enhance raw primary and secondary staining and illumination in microscopic images. The dataset is intorduced to SMOTE (Synthetic Minority Over-sampling Technique) technique to counter over-sampling and under- sampling problems.Processed dataset with a total of 31000 images, is further split to 70% as train data and 30% for test data. Utilizing Watershed algorithm to separate White blood cell images from Red Blood Cells and Plasma. The extracted image is provided as an input to VGG16, a model for instance segmentation, is later utilized to result in separation of single WBC nuclei, and discriminative features extraction (e.g., Area, Perimeter, Solidity, Aspect Ratio and Extent along with features from deep learning such as spatial arrangements and high-level features). Images whose data are retrieved from feature are further presented into the LightGBM model so that cells could be classified into their respective subtypes. The model is tested against standard performance measurement metrics, and we mostly employ the F1 measure to measure the class imbalance that is common in medical data sets.

CHAPTER 2

LITERATURE SURVEY

This Section covers some research articles by various researchers who worked on classification of white blood cells, identification of cells which used machine learning and deep learning model have been collected and done literature review from various Journals and nearly 23 documents are eligible for literature review.

2.1 THEORETICAL INVESTIGATIONS

Chiranjiba et al. (2) propose a deep learning method to enhance COVID-19 diagnosis using chest X-ray (CXR) images. The traditional methods of PCR and CT scan are time and cost-consuming, and therefore the CXR-based deep learning methods are faster alternatives. Current models are not able to learn spatial context in images and are therefore not effective. Following this, the study incorporates an attention module in VGG-16 to enable improved localization of important regions in lung X-rays. It was validated on three available data sets (D1, D2, D3) under 70:30 train-test with accuracies of 79.58%, 85.43%, and 87.49%, respectively, performing better than models such as Inception-ResNetV2, DenseNet-121, and EfficientNet. Its high performance was validated using precision, recall, F-score, and confusion matrices, mainly for the identification of COVID-19 cases.

The research focuses on the advantage of fine-tuning convolution layers and applying the attention mechanism to enhance classification. The proposed model also has fewer training parameters that lower computation costs without sacrificing accuracy. Improvements considered in the future by the authors are improvement with data using GANs and using other pre-trained networks but with lower filter sizes as means of enhanced performance. Due to its enhanced sensitivity and enhanced classifying precision, the proposed model demonstrates an insightful automatic diagnostic mechanism for diagnosing COVID-19 from CXR images in order to accelerate the diagnosis process in a lower-priced manner.

The research article by Nalla praveen et al. (3) uses YOLOv3 technique to identify and distinguish different types of white blood cells (lymphocyte, monocyte, neutrophil and eosinophil). This research work uses a Publicly available Blood cell count dataset (BCCD) which consists of 364 images with labels in RGB format which are of size 320×240×3 in JPEG format. YOLO uses Convolutional neural network to recognize multiple object in single image. YOLOv3 contains 53 Convolutional layers and for detection another 53 is used and totally 106 layers are applied to the image to extract each object in image. Outputs with good accuracy are obtained in their work. This model is classifying WBC with 90% accuracy with considering evaluation metrics such as precision, recall and F1 score. This work can be developed in future by using latest YOLO algorithm.

The paper (8) describes a systematic approach to classifying malaria cell images using deep learning architectures, which include preprocessing, feature extraction, and evaluation. The dataset applied includes parasitic and non-parasitic cell images from Kaggle, which are preprocessed for quality improvement and resized for input into CNNs. They have uses several CNN techniques like ResNet50, DenseNet201, AlexNet, VGG19, GoogleNet, and Inceptionv3, which extract hierarchical features of cells for further classfication.

The processed images are further passed to CNN models that have classified them under parasitic and non-parasitic categories. The architectures' efficiency is then tested using various metrics such as accuracy, precision, and recall. The study establishes that the Gaussian filter significantly enhances the classification performance. DenseNet201 achieved an accuracy of 97.83 percentage. Results highlighted the importance of pre-processing techniques and model selection in providing robust and reliable outcomes for medical image analysis. Deep learning provides utility in automation of malaria images classification, forming a valuable framework for further studies in medical diagnostics. We can see some similar architecture and seen as an developed model in work by Mujahid et al.(9) which uses EfficientNetB2 for accurate image classification and shows a immense accuracy of 97.57% with running 15 epochs using Adam optimizer.

(10) Uses a proposed Nuclei R-CNN technique an advanced instance segmentation method designed to tackle nuclei segmentation challenges in histopathological images. It builds on Mask

R-CNN with enhancements in network architecture, training strategy, and preprocessing. In this method, they start data extraction using ResNet101 in a Feature Pyramid Network (FPN). A Region Proposal Network (RPN) then predicts nucleus-containing regions, which are refined using Non-Maximum Suppression (NMS) and segmented into binary masks via a Fully Convolutional Network (FCN). The output is resized back to its original scale in post-processing.

To handle high-resolution histopathological images, a Block-Merging Algorithm divides images into blocks and intelligently merges them to prevent over-segmentation. In this work, they used preprocessing techniques which includes geometric transformations, blurring of images and K-means clustering to enhance performance of segmentation. This work outperforms U-Net, Dcan and Stardist models in nuclei segmentation achieving high precision and F1 score with compare to other models. This also gives a reliable solution for data extraction for our research work in WBC cell analysis.

CHAPTER 3

METHODOLOGY

3.1 DATA COLLECTION

The white blood cell image segmentation and analysis is from dataset collected from Kaggle. The dataset contains more than 16,600 images for different classes of white blood cells. Preliminary checks on the data revealed that there was a lack of balance in the distribution across the classes: neutrophils in abundance but basophils grossly underrepresented. To correct this imbalance without the use of the traditional oversampling or undersampling methods, SMOTE (Synthetic Minority Over-sampling Technique) was applied which generates new synthetic samples by interpolating between the minority class examples, hence maintaining the natural distribution and preventing overfitting. Images were labelled systematically by their respective class, and class frequency counts were determined. Seeing the extremely high imbalance, SMOTE technique was applied to feature vectors obtained from images. This ensured that all classes of white blood cells would be represented more evenly, a key consideration for the next machine learning operations. Synthetic samples were created only for minority classes and hence did not alter the original feature space.

3.2 OTSU AND WATERSHED ALGORITHM

After acquiring a more even dataset, the focus was turned towards preprocessing the images. All images were first converted to grayscale. Color information must be discarded when the purpose is to highlight structural features rather than color change. Conversion to grayscale not only simplifies subsequent calculations but also enhances contrast between background and cells. The Otsu algorithm was utilized to convert the grayscale images to binary images. Otsu's algorithm calculates an optimal threshold by minimizing intra-class intensity variance, and the produced binary image has cell boundaries become very salient.

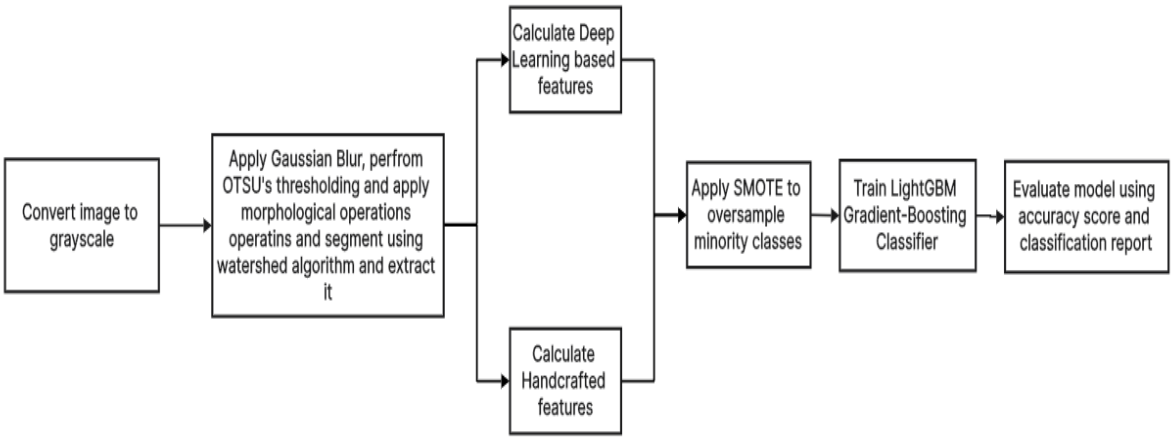


Figure 3.1: Workflow of proposed Model

With the binary images, we proceeded to segmentation with the watershed algorithm. The watershed algorithm operates on the grayscale image as a topographic surface, where pixel intensity is equivalent to elevation. The algorithm begins by identifying markers – local minima that are intended to be the center of cells. The markers are then used as seed points, from which the algorithm 'fills in' around them, essentially marking boundaries where cells are in contact with each other or overlapping.

Segmentation is thus output in a collection of contours that actually mark the edge of each cell. This is an important step in separating cells from the background, thus enabling more advanced morphological analysis. Segmentation was confirmed by comparing areas of segmented cells with a ground truth set of manually annotated images. Quantitative metrics such as Dice coefficient and Jaccard index were calculated to find the overlap between ground truth annotations and automatic segmentation. This evaluation provided us with information about the robustness and accuracy of the model's approach. Combining SMOTE for data balancing with sequential use of Otsu and watershed algorithms for image segmentation gives a robust pipeline that not only functions with imbalances in data but also enhances the accuracy of cell isolation.

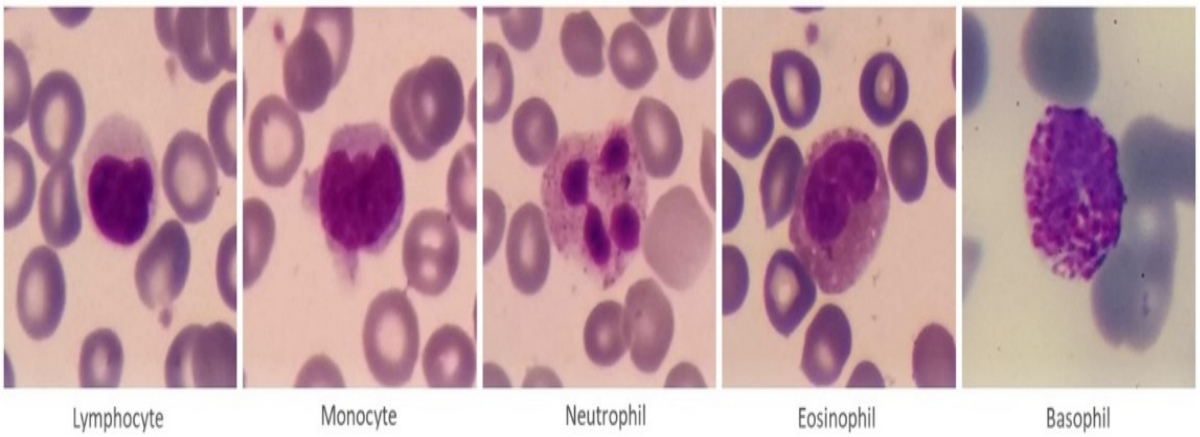


Figure 3.2: Sample of input images for all classes of White Blood Cells

3.3 VGG16 MODEL

VGG16 is a 16-layer deep convolutional neural network with several convolutional layers followed by a few fully connected layers. VGG16 is pre-trained on ImageNet, a large dataset of millions of images from diverse classes. Hence, it is highly proficient in feature extraction for image classification tasks. Through transfer learning, we utilize the pre-trained weights of VGG16 to learn deep spatial features from white blood cell (WBC) images. The network learns and remembers fine details like edges, texture, shapes, and complex structures, which play an important role in distinguishing diverse types of WBCs. Although the bottom layers of VGG16 recognize general features like edges and textures, deeper layers learn task-dependent representations. We fine-tune the final fully connected layers to our dataset so that the feature learning is beneficial for the WBC classification.

In addition to deep learning-based features, we also extract manually crafted morphological features to supplement the automated feature extraction. Handcrafted attributes are area, perimeter, solidity, aspect ratio, and extent, which offer further information regarding WBC morphology. Area and perimeter define the cell's overall size and boundary, solidity is the WBC area divided by its convex hull, aspect ratio differentiates elongated vs. rounded cells, and extent measures the ratio of the cell area to its bounding box. These attributes strengthen the performance of the model for WBC classification based on deep features learned as well as morphological

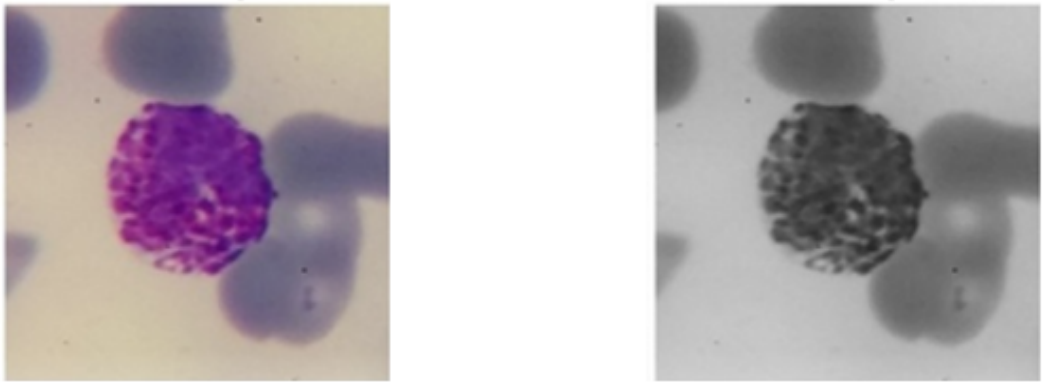


Figure 3.3: Grayscale conversion using watershed algorithm

features domain-specific, making it a more robust system of classification.

3.4 LIGHTGBM BOOSTER

After extracting deep learning features (VGG16) and hand-designed morphological features, they are combined into an overall feature vector and forwarded to LightGBM (Light Gradient Boosting Machine) for classification. LightGBM is an efficient tree-based boosting algorithm that constructs trees leaf-wise rather than level-wise to enable it to make more impactful splits and maximize accuracy with lower computational cost. LightGBM is best designed to work with high-dimensional data like the combined deep and handcrafted feature space, and is highly suited for medical datasets in which class imbalance occurs. LightGBM assigns greater weight to underrepresented WBC classes to promote equitable classification of all cell types. The model learns essential discriminative features from both feature sets and classifies WBCs effectively into classes like Neutrophils, Lymphocytes, Monocytes, Eosinophils, and Basophils.

Input images are preprocessed during prediction to increase feature visibility and minimize noise before being fed into the model. The image is initially transformed to grayscale for the

removal of unwanted color data. Secondly, Otsu's thresholding is used for the separation of the foreground (WBC) and the background so that the contrast between the WBC and its surroundings may be enhanced. Watershed algorithm is then used for the improvement of segmentation through the unambiguous segmentation of the WBC from the background overlap. The segmented image of the WBC is utilized as input to the VGG16 model to yield deep features, and the handcrafted features are calculated concurrently. The resulting hybrid feature set is then passed to the trained LightGBM classifier to calculate a probability score for each WBC class.

By combining VGG16's powerful deep feature extraction, human-designed feature analysis, and LightGBM's effective decision-making, this hybrid system offers a highly efficient, accurate, and independent classification system. This methodology enhances the precision of WBC classification in clinical diagnosis and autonomous blood analysis and delivers a high-performance solution to realistic medical practice.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 OVERVIEW

The classification of White Blood Cells (WBCs) into five categories—Basophil, Neutrophil, Monocyte, Eosinophil, and Lymphocyte was performed using a VGG16-LightGBM hybrid model, achieving an overall accuracy of 87%. For comparison, other deep learning models were also evaluated: Inception ResNetV2, DenseNet, and EfficientNet, with accuracies of 79.58%, 85.43%, and 86.49%, respectively. The dataset initially contained a total of 10,785 images, with 4,339 designated for testing and the remaining used for training. To enhance class balance, Synthetic Minority Over-sampling Technique (SMOTE) was applied, expanding the dataset to approximately 31,000 images. These results highlight the effectiveness of the VGG16-LightGBM model in WBC classification, with a detailed performance analysis provided below.

4.2 VISUAL REPRESENTATION OF CLASSIFIED WHITE BLOOD CELLS (WBC)

The following figure 4.1 shows the images of 5 WBC class classified by our Hybrid Model (VGG16-LightGBM). These images provide the visual confirmation of our model's ability to distinguish between different cell classes.

Basophil: Characterized by a multi-lobed nucleus and large, dark-staining granules.
Neutrophil: Displays a multi-lobed nucleus with fine granules in the cytoplasm.
Monocyte: Features a large, kidney-shaped nucleus with a relatively clear cytoplasm.
Eosinophil: Identified by a bi-lobed nucleus and large, orange-staining granules.
Lymphocyte: Recognized by a large, round nucleus occupying most of the cell volume with a thin rim of cytoplasm.

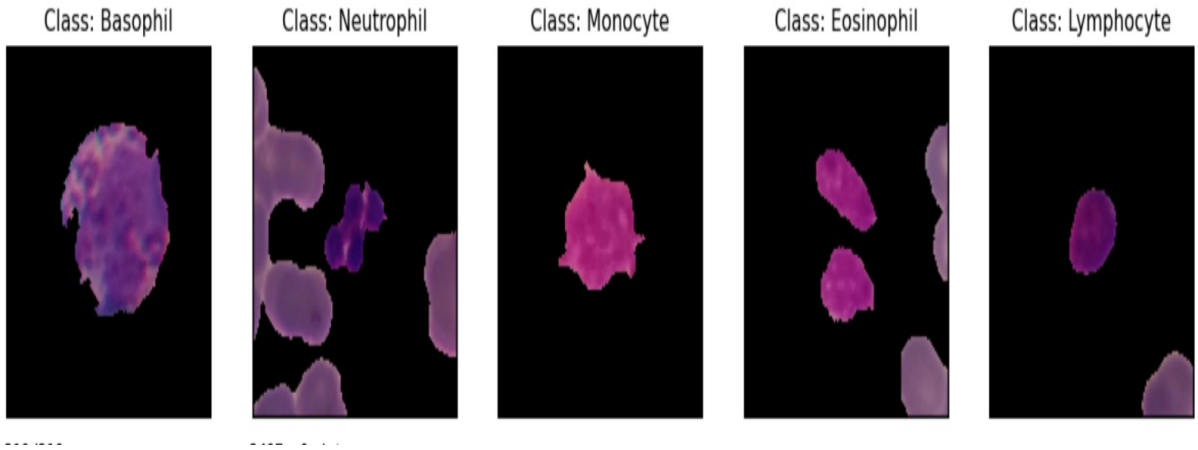


Figure 4.1: Segmented Image

4.3 EXTRACTED FEATURES

Handcrafted features were extracted from the images, including geometric properties such as area, perimeter, solidity, aspect ratio, and extent. These features were calculated for the first five images and are summarized in table 4.3 below:

Image Name	Area	Perimeter	Solidity	Aspect Ratio	Extent
Basophil	4095.00	51.14	0.97	10.3	0.73
Neutrophil	3371.50	220.7	0.98	0.94	0.75
Monocyte	2907.50	204.07	0.98	1.00	0.73
Eosinophil	4338.50	317.62	0.88	0.63	0.72
Lymphocyte	3069.00	214.31	0.97	0.94	0.71

Table. 4.3. Handcrafted features for extracted image

These features highlight the morphological differences between WBCs. For example, the lower aspect ratio (0.63) in Image 4 (likely an Eosinophil) reflects the elongated shape typical of this class, while the high solidity (0.98) in Image 3 (possibly a Monocyte) indicates a compact shape. These handcrafted features, combined with deep learning features, were used for classification in the VGG16-LightGBM model.

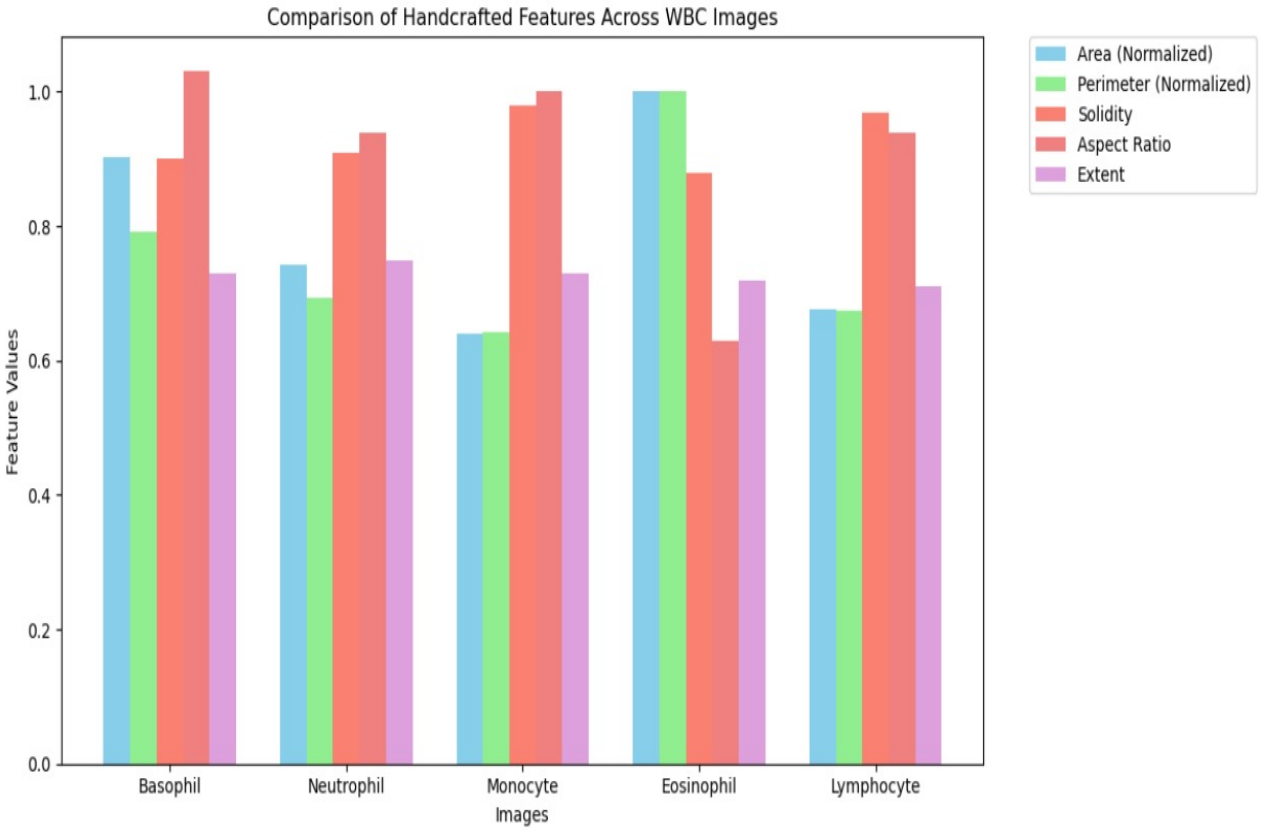


Figure 4.2: Feature extraction comparison between classes

4.4 CLASSIFICATION PERFORMANCE OF OUR HYBRID MODEL

The performance of the VGG16-LightGBM model was evaluated using precision, recall, F1-score, and support metrics for each class. The overall accuracy was 87%, as reported in the output. The detailed classification report is presented in Table 4.4.1.

4.4.1 KEY OBSERVATIONS

- **Basophil (Class 0):** Achieved a high precision of 0.99 and an F1-score of 0.89, indicating strong performance in identifying Basophils, likely due to their distinct morphological features such as large, dark-staining granules.

- **Neutrophil (Class 1):** Recorded the lowest performance with a precision of 0.59, recall of 0.42, and F1-score of 0.49. This suggests that the model struggles to correctly classify Neutrophils, possibly due to overlapping features with other classes like Eosinophils.
- **Monocyte (Class 2):** Demonstrated a balanced performance with an F1-score of 0.88, reflecting the model's ability to correctly identify Monocytes, likely due to their distinct kidney-shaped nucleus.
- **Eosinophil (Class 3):** Showed moderate performance with an F1-score of 0.71, indicating that the model can identify Eosinophils reasonably well, though some misclassifications may occur due to similarities with other classes.
- **Lymphocyte (Class 4):** Achieved the highest F1-score of 0.91, reflecting the model's strong performance in identifying Lymphocytes, likely due to their distinct large, round nucleus.

Class	WBC type	Precision	Recall	F1-score
Class 0	Basophil	0.99	0.81	0.89
Class 1	Neutrophil	0.59	0.42	0.49
Class 2	Monocyte	0.85	0.91	0.88
Class 3	Eosinophil	0.74	0.67	0.71
Class 4	Lymphocyte	0.90	0.92	0.91

Table. 4.4.1. Evaluation metrics for the proposed model

4.4.2 OVERALL METRICS

The overall accuracy of 87% indicates that the VGG16-LightGBM model performs well across all classes, correctly classifying a significant majority of the 4339 test samples. This high accuracy underscores the model's reliability for practical applications in medical diagnostics, where accurate WBC classification is crucial.

4.5 COMPARATIVE STUDY WITH OTHER MODELS

The performance of the Hybrid (VGG16-LightGBM) model was compared with other deep learning models: **Inception ResNetV2**, **DenseNet**, and **EfficientNet**. The accuracies of these models are summarized in Table 4.5 below.

Model Name	Accuracy %
Inception ResNetV2	79.58 %
DenseNet	85.43 %
EfficientNet	86.49 %
VGG16-LightGBM (Our Proposed Model)	87 %

Table. 4.5. Performance Comparison of various Models

- **VGG16-LightGBM (87%):** Outperformed Inception ResNetV2 and DenseNet, and EfficientNet. The hybrid approach of combining VGG16 for feature extraction with LightGBM for classification likely contributed to this performance, as it leverages both deep learning and handcrafted features, providing a more comprehensive representation of the WBCs.

- **EfficientNet (86.49%):** Achieved a very close accuracy to VGG16-LightGBM, indicating that EfficientNet's architecture, which is designed to scale efficiently across depth, width, and resolution, is highly effective for this task. However, it slightly underperformed compared to VGG16-LightGBM, possibly due to the lack of handcrafted features, which provided additional discriminative power in the hybrid model.

- **DenseNet (85.43%):** Performed well but lagged behind VGG16-LightGBM and EfficientNet. DenseNet's strength lies in its dense connectivity, which promotes feature reuse, but it may not have captured the same level of detail as VGG16-LightGBM's combined feature set.

- **ResNetV2 (79.58%):** Recorded the lowest accuracy among the models. While Inception ResNetV2 combines the benefits of Inception modules and residual connections, its lower performance suggests that it may not be as well-suited for this specific dataset, possibly due to less effective feature extraction for WBC classification.

4.6 DISCUSSION AND FUTURE SCOPE

The detailed performance analysis and comparison with other models provide insights into the model's strengths and areas for improvement.

4.6.1 FEATURE CONTRIBUTION

- The handcrafted features listed in table 4.3 show variability across images, which likely helped the model differentiate between classes. For example, the lower aspect ratio (0.63) in Image 4 (likely an Eosinophil) reflects the elongated shape typical of this class, while the high solidity (0.98) in Image 3 (possibly a Monocyte) indicates a compact shape.

- The combination of these handcrafted features with deep learning features extracted by VGG16 provided a comprehensive representation of the WBCs.

4.6.2 CLASS-WISE PERFORMANCE INSIGHTS

- The high F1-scores for Lymphocytes (0.91) and Monocytes (0.88) indicate that the model excels at identifying these classes, likely due to their distinct morphological features, such as the large, round nucleus of Lymphocytes and the kidney-shaped nucleus of Monocytes.

- The lower F1-score for Neutrophils (0.49) suggests that the model struggles with this class, possibly due to overlapping features with Eosinophils, such as multi-lobed nuclei, which may lead to misclassifications.

- Basophils and Eosinophils, with F1-scores of 0.89 and 0.71, respectively, show good to moderate performance, indicating that the model can identify these classes reasonably well, though there is room for improvement.

4.6.3 STRENGTHS

- **High Accuracy:** The accuracy which we got is promising for a multi-class classification problem, especially in a medical context where precise identification of WBCs is critical.
- **Feature Variability:** The extracted features, such as area, perimeter, and solidity, captured morphological differences between WBCs, aiding in classification.
- **Superior Performance:** The VGG16-LightGBM model outperformed Inception ResNetV2, DenseNet, and EfficientNet, highlighting the effectiveness of the hybrid approach.

4.6.4 LIMITATIONS

- **Class-specific Challenges:** The lower performance on Neutrophils (F1-score of 0.49) indicates that the model struggles with this class, possibly due to overlapping features with other classes like Eosinophils.
- **Generalization:** The model's performance on unseen data remains uncertain, as the results are based on the provided dataset. Further validation on external datasets is necessary to ensure generalizability.

4.6.5 FUTURE IMPROVEMENTS

- **Ensemble Methods:** Combining the VGG16-LightGBM model with other classifiers, such as Random Forest or XGBoost, in an ensemble framework could enhance overall performance by leveraging the strengths of multiple models.
- **Multi-modal Learning:** Extending the model to incorporate multi-modal data, such as combining WBC images with other imaging modalities (e.g., flow cytometry data or genetic markers), could improve classification accuracy by capturing a broader range of diagnostic features.
- **Cross-domain Applications:** Adapting the model for other medical image classification tasks, such as identifying cancer cells or classifying other types of blood cells (e.g., red blood cells, platelets), could broaden its applicability in the field of medical diagnostics.

CHAPTER 5

CONCLUSION

In this research, we have proposed a hybrid deep learning framework for the classification of white blood cells (WBC) and the extraction of essential features using an integrated approach. The framework consists of three stages: first, the application of the watershed algorithm to accurately segment and extract WBC regions; second, the use of VGG16, a pre-trained convolutional neural network (CNN), to extract critical morphological features from the segmented regions; and finally, the classification of WBC types using LightGBM, a gradient-boosting framework that ensures efficient and accurate classification. By utilizing these advanced models, our approach achieves a high accuracy of 87%, enabling the precise extraction of crucial features that facilitate reliable disease diagnosis.

The proposed method not only improves the accuracy of feature extraction but also enhances efficiency in detecting cells, extracting features, and classifying them, thereby reducing the subjectivity and time involved in manual examination. Additionally, by incorporating preprocessing techniques such as image normalization and contrast enhancement, the framework optimizes model performance and ensures reliable analysis across various histopathological images.

Future work can focus on enhancing accuracy by exploring more advanced segmentation algorithms and integrating multi-model approaches to improve feature representation. Moreover, using larger and more diverse datasets can refine model generalization, while fine-tuning the framework with techniques such as transfer learning and hyperparameter optimization can further improve classification accuracy and accelerate the training process. This improved framework has the potential to greatly enhance automated hematological analysis and improve clinical decision-making.

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